Heart Disease Prediction with Logistic Regression, Regularization, and PCA

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I. Introduction and Background

According to the Centers for Disease Control and Prevention (2024), heart disease is one of the leading causes of death in the United States and places a substantial strain on healthcare systems. Early detection of individuals who may be at risk is essential for enabling timely interventions, improving treatment outcomes, and reducing mortality rates. With increasingly available medical datasets, data science methods provide powerful tools for analyzing clinical information and uncovering patterns that may help predict heart disease risk.

This project analyzes a dataset containing clinical and demographic variables related to heart disease from the UC Irvine Machine Learning Repository to investigate predictive performance and relationships within the data. The goal is to identify which variables are most strongly associated with heart disease risk and to evaluate how accurately heart disease can be predicted using individual patient diagnostic information under different modeling approaches. In particular, we compare the predictive performance and computational efficiency of a baseline logistic regression model with Lasso- and Ridge-Regularized models. We also explore the use of dimensionality reduction through Principal Component Analysis (PCA) to assess whether reducing feature dimensionality can improve predictive performance. Model performance is evaluated using ROC-AUC and classification accuracy, and efficiency is assessed through model runtime.

II. Dataset Description

The dataset contains information for patients evaluated for heart disease. Each observation represents an individual patient, and the response variable indicates the presence or absence of heart disease. The dataset includes 13 predictor variables describing patient characteristics, diagnostic test results, and medical measurements that may be associated with heart disease risk.

Table 1. Predictors and descriptions

Predictor	Description
age	Age of patient in years
sex	Biological sex
cp	Chest pain type
trestbps	Resting blood pressure
chol	Serum cholesterol (mg/dl)
fbs	Fasting blood sugar > 120 mg/dl
restecg	Resting electrocardiographic results
thalch	Maximum heart rate achieved

exang	Exercise induced angina (chest pain caused by reduced blood flow to the heart during exercise)
oldpeak	ST depression induced by exercise relative to rest (a measure from an ECG indicating potential heart stress)
slope	Slope of the pearl exercise ST segment
ca	Number of major vessels coloured by fluoroscopy (indicates blockage)
thal	Thalassemia test result (indicates blood disorder affecting hemoglobin production)

III. Data Preprocessing

Before moving onto analysis, some preprocessing steps were applied. The original dataset uses a categorical target variable where a value of 0 indicates that there is no disease, while a value of 1-4 indicates different levels of the disease. For the purposes of simplicity and interpretability, target values 1-4 were combined to form a binary target variable, where 0 represents the absence of disease and 1 indicates the presence of disease.

The column with the original encoding for the target variable was dropped along with the ID column. Additionally, since there was a lot of missing information in many of the locations of data collection, such as missing fluoroscopy information at the Switzerland, , Hungary, and Long Beach locations, the dataset was filtered to only include information from the Cleveland location. All remaining observations with null values were dropped. This left 297 observations total for modeling. Prior to modeling, dummy variables were created for all categorical predictors and standardization was applied.

IV. Exploratory Data Analysis

Exploratory data analysis (EDA) was conducted to better understand the distribution of variables in the dataset and to identify potential relationships between the predictors and the presence of heart disease. This step also helps assess whether variables satisfy assumptions for modeling and motivates the use of several techniques.

4-1. Distribution of Predictors

To examine the distribution of the predictors, exploratory visualizations were created. Box-and-whisker plots were used for continuous predictors to compare their distributions across heart disease outcomes (Figure S1, Supplementary Materials). These plots indicate that several variables do not exhibit clear separation between the two classes. For example, cholesterol and resting blood pressure show similar distributions for patients with and without heart disease, suggesting that these predictors may have limited discriminative power individually.

For categorical predictors, bar charts were created to display the frequency for each category across outcome classes (Figure S2, Supplementary Materials). Some features, such as

fasting blood sugar and resting ECG, appear relatively balanced between the two groups, indicating they may not be strong predictors in a pairwise setting.

4-2. Outcome Class Balance

The class distribution of the outcome variable was also examined. Balance across outcome classes is desirable to prevent the model from overfitting to the majority class. In this case, the two outcome classes, the absence and presence of heart disease, are fairly balanced, each representing approximately 54% and 46% of the data respectively. As such, strategies to handle imbalance, such as resampling and threshold tuning, were not considered.

4-3. Pairwise Correlation

Pairwise relationships between variables and the heart disease outcome were further examined using appropriate correlation measures based on variable type. For continuous predictors, Pearson correlation coefficients were calculated to evaluate the strength and direction of linear relationships with heart disease. For categorical predictors, Cramér's V was used.

The results indicated that cholesterol, resting blood pressure, fasting blood sugar, and resting ECG exhibited relatively weak correlations with heart disease. The full set of pairwise correlation coefficients are provided in the Supplementary Materials (Table S1). It is important to note that weak pairwise associations do not necessarily imply that these predictors are uninformative in a multivariate context. Variables that appear weak individually may still contribute meaningful information when considered jointly with other predictors.

This observation motivates the use of modeling approaches that evaluate predictors simultaneously. In particular, Lasso-regularized logistic regression is used to perform joint feature selection, while PCA-based logistic regression is considered to assess whether lower-dimensional combinations of predictors can improve predictive performance.

4-4. Multicollinearity

To evaluate multicollinearity among predictors, variance inflation factors (VIF) were computed, where high VIF values indicate that a predictor may be strongly correlated with the other variables in the model. For this dataset, most VIF values were small and indicate little evidence of multicollinearity. However, a few predictors show relatively higher VIF values of 5-6. Though this could be due to the fact that some of these predictors are dummy variables derived from the same categorical feature, which introduces linear dependencies, this still motivates the usage of Ridge-regularized logistic regression to evaluate whether multicollinearity meaningfully impacts predictive performance. The full set of VIF values for all predictors is provided in the Supplementary Materials (Table S2).

Additionally, a heatmap of the correlation matrix for continuous predictors is provided in the Supplementary Materials (Figure S3), showing moderate negative correlations between some predictors.

4-5. Linearity Assumption Check

Finally, the relationship between continuous predictors and the log-odds of heart disease was examined. Assessing these relationships helps evaluate whether the predictors satisfy the linearity assumption required for logistic regression models. The relationship between most continuous predictors and the log-odds of heart disease are approximately linear, so transformations were not considered for them. Some non-linearity can be observed in the relationship between resting blood pressure and the log-odds of heart disease, so a polynomial transformation was considered for this variable. However, since it did not improve model performance, in the interest of interpretability, we retained the linear specification.

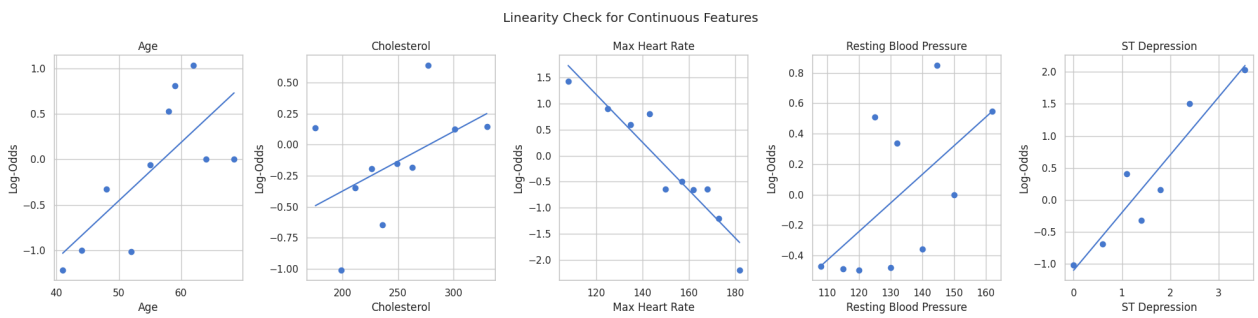

Figure 1. Plot of log-odds of outcome against continuous predictors

V. Methods

5-1. Cross-Validation Procedure

To evaluate model performance and avoid overfitting, we implemented 5-fold stratified cross-validation. Stratification makes sure that each portion of patients with and without heart disease remains proportionate across each fold. In each fold, the data is split up into 80% training data and 20% test data. The model is then trained on the training fold and then the model is used to predict patient categorization on the validation fold. Performance is then evaluated using ROC-AUC and classification accuracy, with probabilities converted to class labels using a threshold of 0.5. This process is repeated for all 5 folds and the final performance is measured with the reports of the mean and standard deviation across the 5 folds which provides a more reliable, generalized estimate of the model's performance.

5-2. Baseline Logistic Regression

As a baseline predictive model, we implemented logistic regression with the standardized feature set. To estimate an unregularized model, the regularization parameter C was set to a large value $C = 10^{12}$ which effectively removes the penalty term. The model was trained with a LBFGS solver which is effective for logistic regression with continuous variables used for predictions. Performance was then evaluated with the 5-fold stratified cross validation, then evaluating the results based on the ROC-AUC and classification accuracy. This baseline model then serves as a reference point for comparing the effects of regularization and dimensionality reduction in future prediction models.

5-3. Ridge-Regularized Regression

We then implemented ridge-regularized logistic regression in order to address potential multicollinearity among the predictors and to reduce any overfitting of data in the prediction model. We use L2 regularization to penalize large model coefficients during model training which prevents the model from relying on one single feature too much. Ridge regression also stabilizes predictors that are correlated with one another. A grid of regularization strengths was evaluated by varying the inverse regularization parameter C across several values ranging from strong to weak regularization. The model performance is then evaluated with the 5 fold stratified cross validation and summarized with the mean and standard deviation of ROC-AUC and classification accuracy across folds.

5-4. Lasso-Regularized Regression

The next model implemented is lasso-regularized logistic regression with L1 regularization. Lasso penalizes the absolute magnitude of the prediction coefficients, which can shrink the coefficients to exactly zero and thus performs automatic feature selection. This allows the model to identify a subset of predictors which contribute the strongest in heart disease risk. The model is trained with a lib-linear solver which supports the L1 regularization in logistic regression. A range of regularization strengths is then evaluated with the inverse regularization parameter C and for each value of C, the model performance and number of selected predictors is recorded with the 5-fold stratified cross-validation.

5-5. Logistic Regression on PCA-Reduced Data

The logistic regression models are then trained with the principal component analysis (PCA) representations of the predictor variables in order to determine whether dimensionality reduction may improve the model performance. PCA transforms the standardized predictors into a smaller set of independent pieces which capture the directions of variance in the data. PCA is applied within each cross validation fold and only on the training portions of the fold in order to

avoid information leakage and avoid overfitting. The logistic regression models were then trained on the reduced feature set. Multiple k values which represent the number of principal components retained were then evaluated in order to find how dimensionality reduction affected the model's performance accuracy. The model is then evaluated based on the ROC-AUC and classification accuracy under the 5-fold stratified cross-validation.

VI. Results

The full set of training and test results for all models are provided in the Supplementary Materials (Table S3). All models were computationally efficient, with training and evaluation completed in milliseconds.

The baseline logistic regression model achieved an average ROC-AUC of about 0.889 with a classification accuracy of about 0.821 (Table S3) under the 5-fold cross-validation. Adding the ridge regularization improved the predictive performance, gaining an AUC of about 0.907 and an improved accuracy of 0.832. This indicates that moderate regularization helps stabilize the model when some predictors show correlation with other predictors.

Lasso regularization produced a similar predictive performance to ridge regression with a test AUC score of 0.897 and an accuracy of 0.821 along with performing feature selection by shrinking some coefficients to zero. Lasso chose 16 of the 18 predictor variables to keep, removing the age variable because age pattern is found with the number of vessels, ST depression, exercise-induced angina, and chest pain type, and then also removing resting ECG (ST-T) because there are few of those observations in the dataset (Table S4). The number of selected predictors varied based on the regularization strength which highlighted a subset of variables that had the strongest association with heart disease risk (Table S4).

Logistic regression models trained on PCA-reduced data showed that moderate dimensionality reduction can maintain and even slightly improve the predictive performance of the model. Among the tested values of k, keeping five dimensions produced the highest AUC of about 0.904 and an accuracy of 0.832 while explaining about 54% of the variance. Increasing the amount of components beyond that point didn't substantially improve the predictive performance which suggests there are weakening returns from adding more components. Overall, ridge regularization produced the strongest predictive performance amongst all the evaluated model approaches.

VII. Discussion

This study examined which clinical and demographic variables are associated with heart disease risk and how different modeling approaches affected predictive performance. Several variables showed strong relationships with heart disease including: thalassemia status, number of major vessels (ca), exercise-induced angina, and ST depression (oldpeak). Age and sex also showed moderate associations with heart disease. Their relationships were presented in the low-dimensional representations captured through PCA in which the variables related to cardiac stress and vessel blockage strongly contributed to the leading directions of variance.

The results also showed heart disease can be predicted with a high accuracy using available patient diagnostic information. The baseline logistic regression model achieved an AUC score of 0.889 which regularization increased the performance to 0.907. Logistic regression trained on reduced features also performed well, the best achieving an AUC of 0.904, which demonstrates there are core predictive variables that strongly contribute to predicting whether a patient does or doesn't have heart disease.

When comparing the dimensionality reduction and supervised learning approaches, ridge regression produced the highest predictive performance while maintaining clear interpretability for variable coefficients. The PCA-based models had a similar predictive accuracy while reducing dimensionality of the feature space, however interpretation became less direct due original variables being combined for predictions. Overall, both approaches produced comparable and competent predictive performances which suggests either method is effective for modeling heart disease risk in this dataset.

VIII. Conclusion

This study compared different regularization approaches and PCA-based dimensionality reduction in logistic regression models for heart disease prediction using clinical and demographic variables. The results showed that heart disease risk can be predicted with high accuracy using available diagnostic information, with ridge regularization producing the strongest performance. The PCA-based model achieved similar predictive accuracy, though at the cost of reduced interpretability. Overall, regularized logistic regression provides a strong balance of predictive accuracy, interpretability, and computational efficiency for heart disease risk prediction.

A key strength of this study is the use of interpretable models, which allow clear interpretation of relationships between clinical variables and heart disease risk. Regularization improved model stability under correlated predictors, and 5-fold cross-validation provided reliable performance estimates. Overall, the models achieved strong predictive accuracy. Despite these strengths, several limitations should be noted. The dataset after preprocessing contains approximately 300 patient observations, which is small and may limit the generalizability of the findings to border populations. Additionally, the use of linear models may fail to capture potential nonlinear relationships between predictors and heart disease risk. In addition, although PCA-based dimensionality reduction helped simplify the feature space, it reduced the interpretability of the predictors because original variables were combined into principal components. Furthermore, clinical datasets may contain measurement noise or bias that can influence model estimates.

Future work could explore nonlinear machine learning models, such as random forests or gradient boosting methods, which may capture more complex relationships in the data. Expanding the analysis to larger clinical datasets would also help validate the robustness and generalizability of the findings. Finally, further feature engineering and the inclusion of interaction terms may improve predictive performance and provide deeper insights into the factors associated with heart disease risk.

References

Centers for Disease Control and Prevention. (2024, October 24). *Heart disease facts*. Heart Disease. https://www.cdc.gov/heart-disease/data-research/facts-stats/index.html

Supplementary Materials

Table S1. Pairwise correlation coefficients for all predictors

Predictor	Type	Correlation	Method
Age	Continuous	0.2271	Pearson
Cholesterol	Continuous	0.0803	Pearson
Max Heart Rate	Continuous	-0.4238	Pearson
Resting Blood Pressure	Continuous	0.1535	Pearson
ST Depression	Continuous	0.4241	Pearson
Thalassemia	Categorical	0.5269	Cramér's V
Chest Pain Type	Categorical	0.5101	Cramér's V
Num Major Vessels	Categorical	0.4934	Cramér's V
Exercise-Induced Angina	Categorical	0.4142	Cramér's V
ST Slope	Categorical	0.3826	Cramér's V
Sex	Categorical	0.2712	Cramér's V
Resting ECG	Categorical	0.1796	Cramér's V
Fasting Blood Sugar	Categorical	0.0000	Cramér's V

Table S2. VIF values of all predictors

Predictor	VIF
Thalassemia: Reversible Defect	4.7957
Thalassemia: Normal	5.4912
ST Slope: Upsloping	5.9076
ST Slope: Flat	4.8403
Exercise-Induced Angina: True	1.4349
Resting ECG: ST-T Abnormality	1.0769
Resting ECG: Normal	1.1215

Fasting Blood Sugar: True	1.1271
Chest Pain Type: Typical Angina	1.2428
Chest Pain Type: Non-Anginal	1.5135
Chest Pain Type: Atypical Angina	1.5058
Sex: Male	1.3422
Num Major Vessels	1.3775
ST Depression	1.8667
Max Heart Rate	1.7357
Cholesterol	1.1506
Resting Blood Pressure	1.2266
Age	1.5028

Table S3. Predictive Model ROC-AUC and Accuracy Results

Model Type	Training ROC-AUC Result	Test ROC-AUC Results	Training Accuracy	Test Accuracy
Baseline Logistic Regression	0.9399	0.8887	0.8754	0.8214
Ridge Regression	0.9215	0.9072	0.8535	0.8316
Lasso Regression	0.9378	0.8968	0.8670	0.8214
Logistic Regression on PCA-Reduced Data	0.9141	0.9041	0.8476	0.8315

Table S4. Predictors Selected by Lasso-Regularized Logistic Regression

Predictor	Coefficient
Num Major Vessels	1.015
Thalassemia: Reversible Defect	0.591
Sex: Male	0.535
ST Depression	0.389

Exercise-Induced Angina: True	0.315
Resting Blood Pressure	0.285
ST Slope: Flat	0.265
Cholesterol	0.114
Chest Pain Type: Non-Anginal	-0.679
Chest Pain Type: Typical Angina	-0.434
Max Heart Rate	-0.315
Chest Pain Type: Atypical Angina	-0.248
ST Slope: Upsloping	-0.192
Resting ECG: Normal	-0.184
Fasting Blood Sugar: True	-0.109
Thalassemia: Normal	-0.060

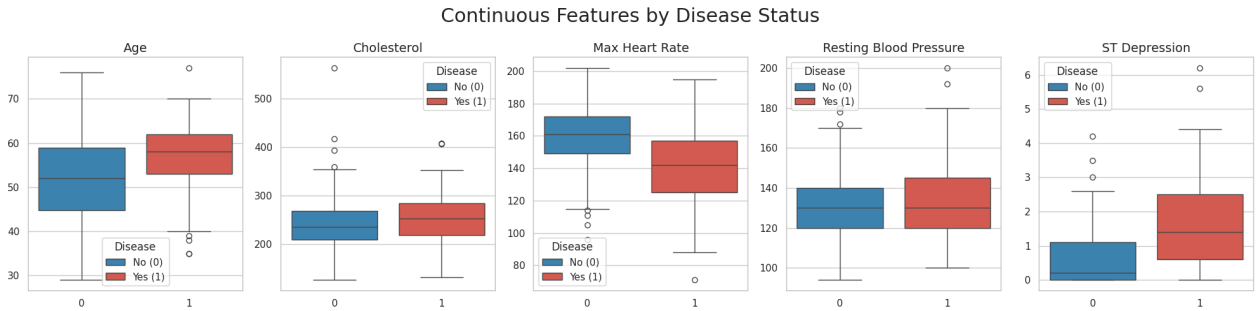

Figure S1. Distribution of continuous variables by heart disease outcome

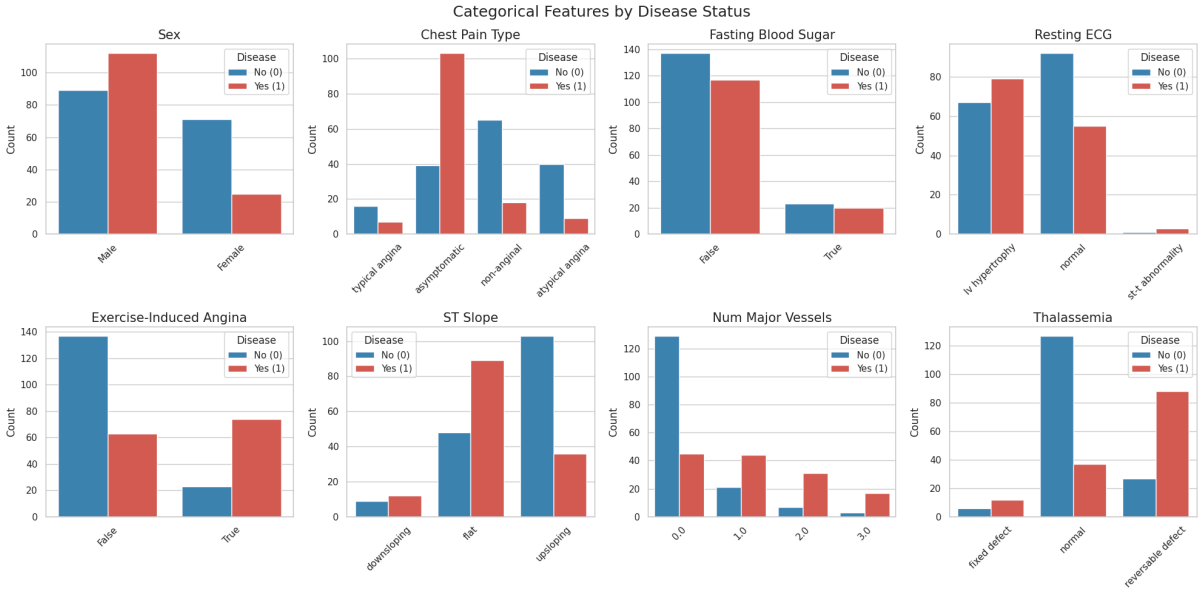

Figure S2. Distribution of categorical variables by heart disease outcome

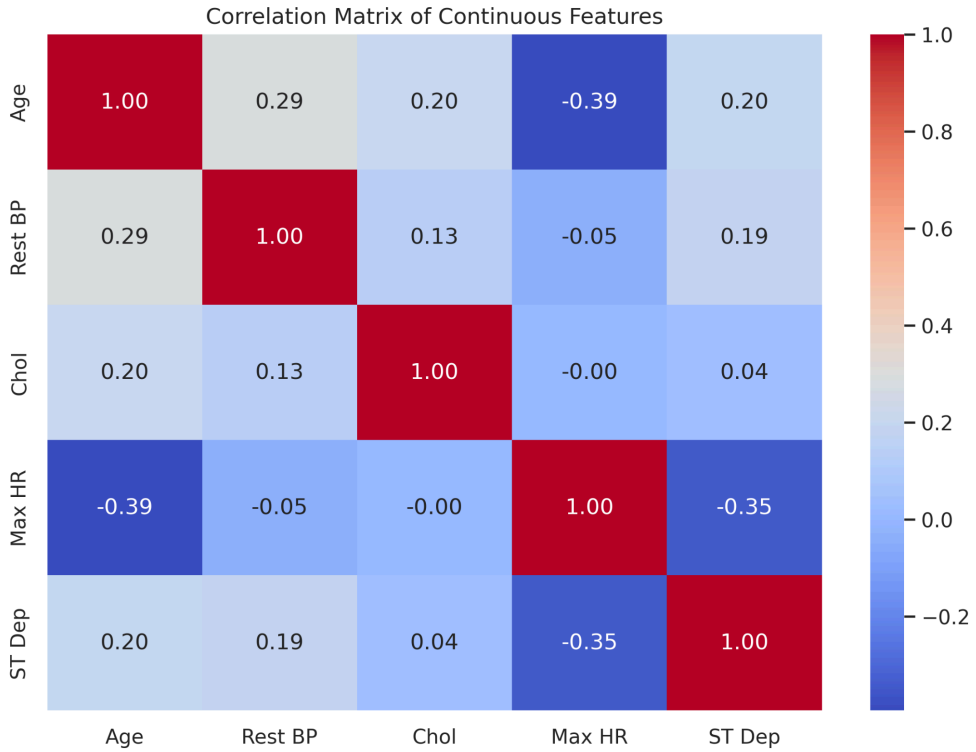

Figure S3. Heatmap of the correlation matrix of continuous features

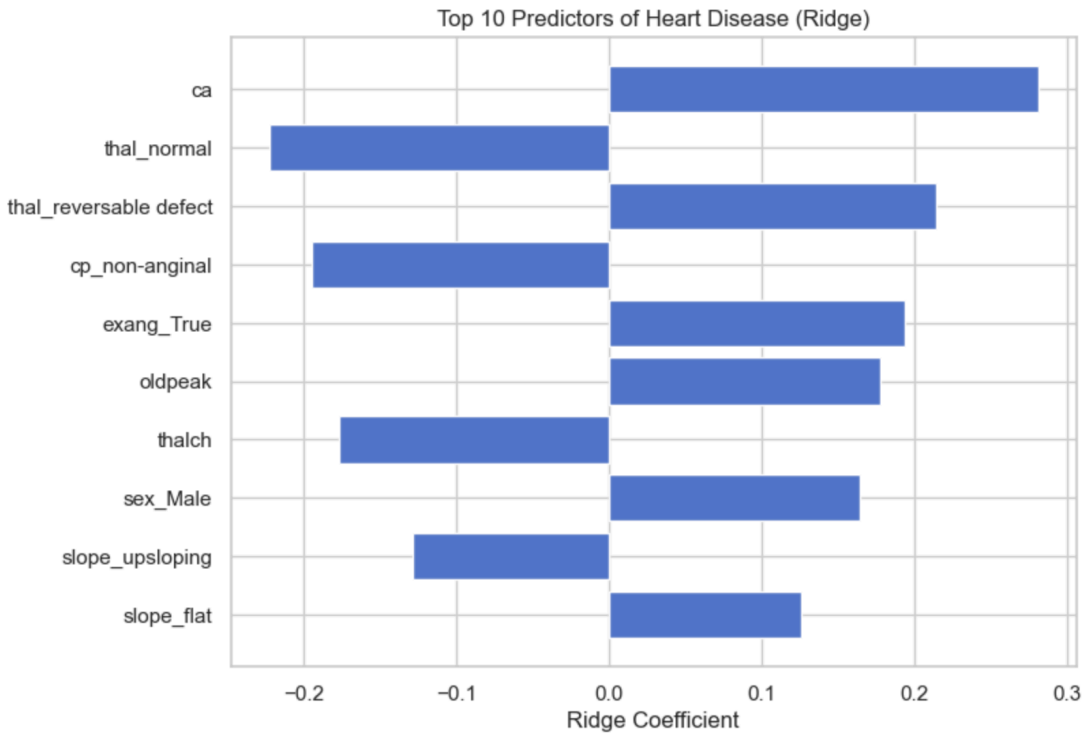

Figure S4. Largest Ridge Logistic Regression Coefficients for Predicting Heart Disease